

Starter



3:00

With a partner, write down as many words about the weather as you can. (you must be ready to explain the meanings of the words you write down)

Be prepared to share them with the class!

Weather and Climate

Success criteria

Lesson Title	Working towards	Working at	Working beyond
What is an Ecosystem	I can name some living (biotic) and non-living (abiotic) parts of an ecosystem.	I can define key terms like community, population, habitat and explain how energy moves through a food web	I can explain the role of decomposers and interpret pyramids of numbers to show energy transfer in ecosystems.
Interactions within Ecosystems	I can identify some habitats found in the Sonoran Desert and other ecosystems.	I can describe interactions between organisms and their environment in different ecosystems.	I can explain the differences between habitats and ecosystems and how organisms are adapted to survive in these settings.
Adaptations and Competition	I can give examples of how plants or animals are adapted to their environments.	I can describe how adaptations help species survive and explain competition between organisms.	I can analyse mutualistic relationships and present research on how different species interact in ecosystems.
Invasive Species	I can name an invasive species and describe what a native species is.	I can describe how invasive species affect native plants and animals in an ecosystem.	I can explain why controlling invasive species is important and evaluate different methods of control.
Case Study - New Zealand (Debate)	I can identify an invasive species problem in New Zealand.	I can describe the ecological impact of invasive species and outline arguments for control.	I can construct and defend an argument in a debate using evidence about species control and conservation.
Bioaccumulation	I can state what an insecticide is and give a simple example.	I can describe how chemicals like DDT build up in organisms and impact the food chain.	I can explain why organisms at the top of the food chain have higher toxin levels and the environmental consequences.
Weather and Climate	I can describe the difference between weather and climate.	I can explain how weather affects plants and animals in an ecosystem.	I can analyse weather data to predict effects on local ecosystems and seasonal changes.
Climate and Atmosphere	I can name key gases in the atmosphere.	I can explain how greenhouse gases affect Earth's climate and describe human impacts.	I can evaluate how climate change may affect ecosystems and propose possible solutions.

Starter:

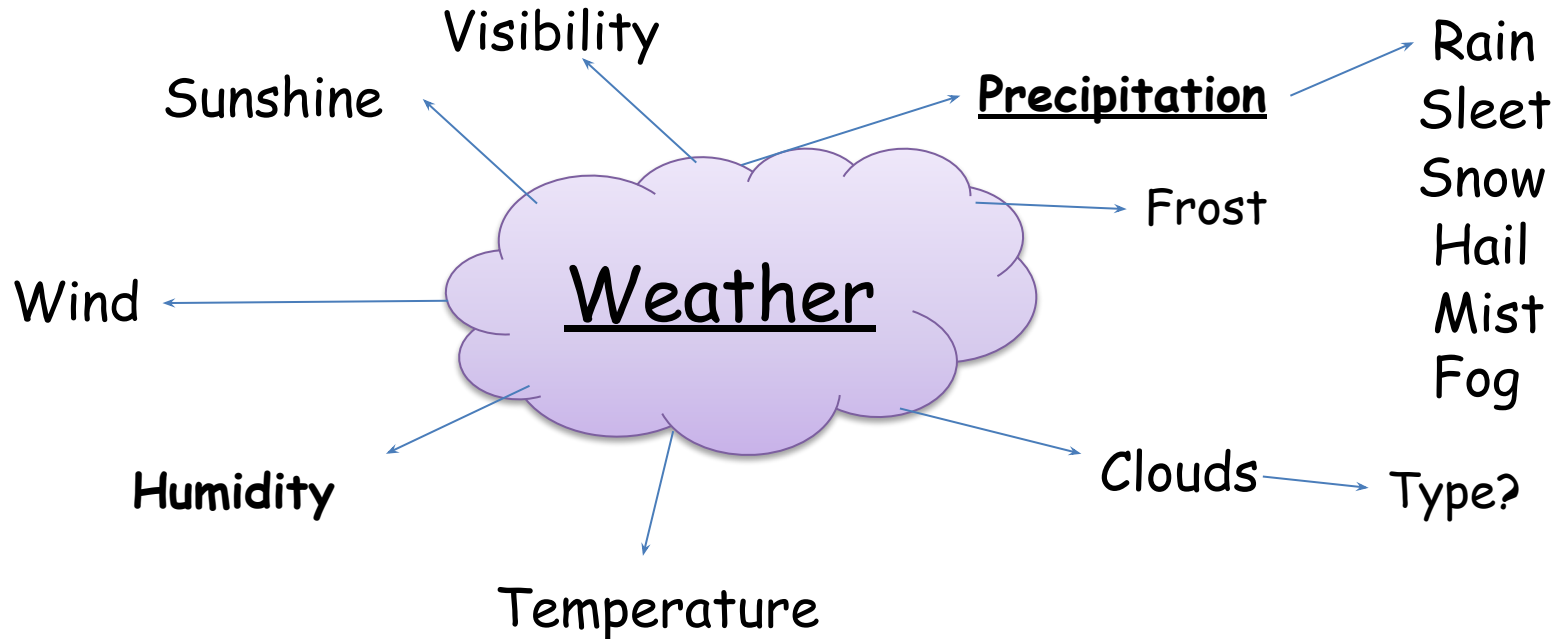
- Describe what the weather is like at this location



Weather:

The day to day changes in the atmosphere

How many types different types weather can you come up with:



Finding out about the weather

In this activity you will find and evaluate information about the weather in a particular place.

- 1 Work with a partner. Choose a place, anywhere in the world. It might be somewhere you would like to go on holiday – a beach, mountains for skiing, or somewhere to sail a boat.
- 2 Find as much information as you can about the weather in that place. For example: What is the average number of hours of sunshine? How much rain is expected? How windy is it? How much snow will there be?

Make sure you find information from different sources. You might look at websites for travel agents, the national weather service or local weather watchers. You could also look at past weather records over a few years and compare them.

- 3 Think about these questions and discuss them with your partner:
 - Is the information you find from all sources exactly the same?
 - Can you suggest why this is?
 - Which source of information do you think is more reliable?
 - Could some sources be biased? Perhaps someone wants to give the best view of the weather to encourage you to go there.



15:00

15 minutes of research and write your findings on the table.

After timer ends, we'll go clockwise to each other's table to read their findings



3:00

What is Climate 气候 / 기후 ?

Think

Pair

Share



Describe the climate of these two locations.



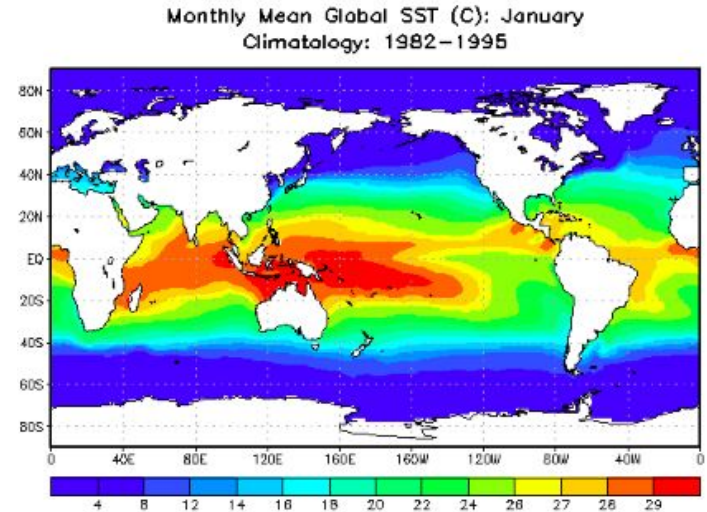
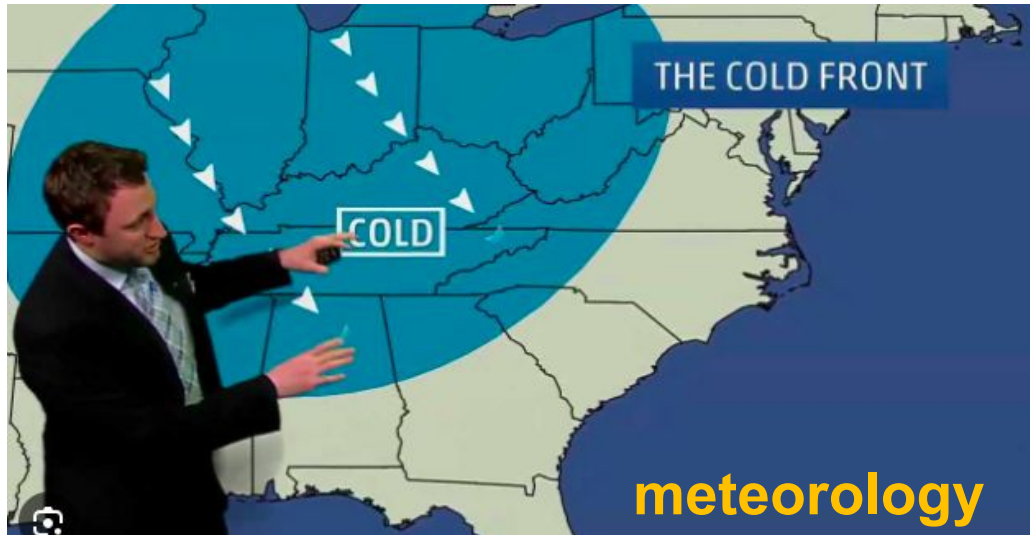
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Terminologies

The study of weather is called **meteorology**

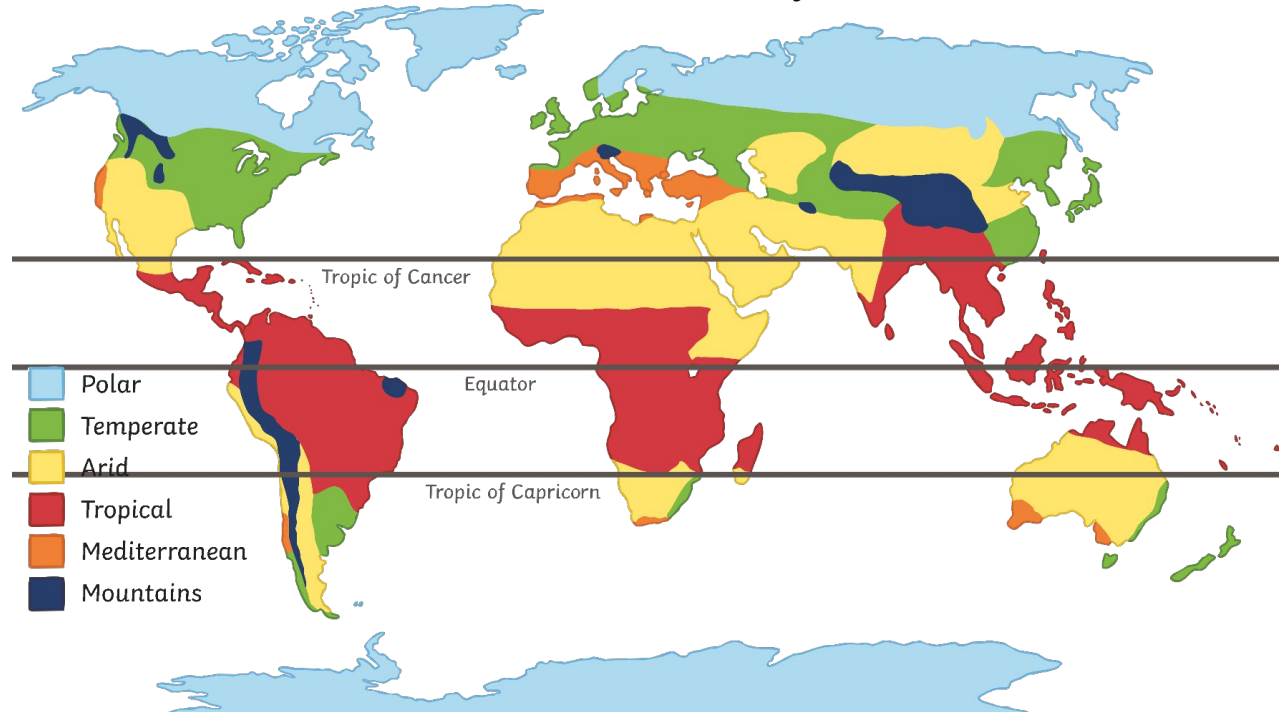
The study of climate is called **climatology**



climatology

World Climate Zones

This is a map of the main world climate zones. A climate zone is an area that has its own climate. Climate zones also have their own type of vegetation and wildlife too! Which climate zone is Malaysia in?



Polar Climate Zone

How would you describe a polar climate?



Polar climates have temperatures which are usually below freezing and can reach -60°C in winter. Polar areas are usually covered by snow and ice throughout the year.

- Very cold and dry all year

Temperate Climate Zone 温带气候 / 온화한 기후

How would you describe a temperate climate?



Temperate climates vary greatly at different times of year, with four distinct seasons.

- Cold winters and mild summers

Mediterranean Climate Zone 地中海气候 / 지중해성

How would you describe a Mediterranean climate? 기후



Mediterranean climates have long, warm, dry summers and wet winters.

- Mild winters and hot, dry summers

Arid Climate Zone 干旱气候 / 건조한

How would you describe an arid climate?

기후



Arid climates lack natural water sources, with little rainfall. They are very dry and hot.

- Hot and dry all year

Tropical Climate Zone 热带气候 / 열대 기후

How would you describe a tropical climate?



Tropical Climates have high temperature rainfall and humidity all year. Some areas may have a wet and dry season.

- Hot and wet all year

Mountains/Tundra/Taiga Climate Zone

山地/苔原/泰加林气候

산/툰드라/타이가 기후

How would you describe a mountain climate?



Mountains have a different climate to their surrounding areas. The temperature on mountains becomes colder the higher the altitude gets. They also tend to have much wetter climates than the surrounding land.

- Very cold all year.

World Climate Zones

20:00

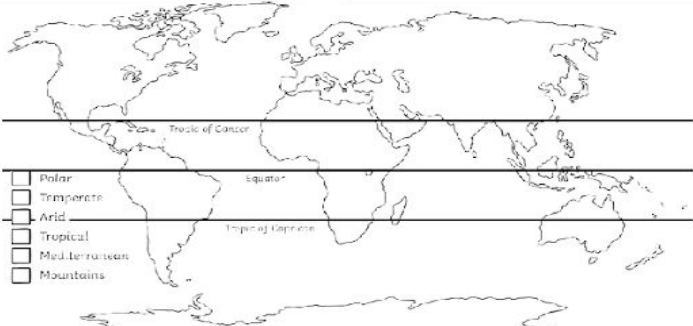
Complete the World Climate Zones Activity Sheet.

Shade each climate zone a **different colour** and **complete the key**.
Write a **short description** of each climate zone in the boxes.

- **Polar**
- **Temperate**
- **Mediterranean**
- **Arid**
- **Tropical**
- **Mountain**



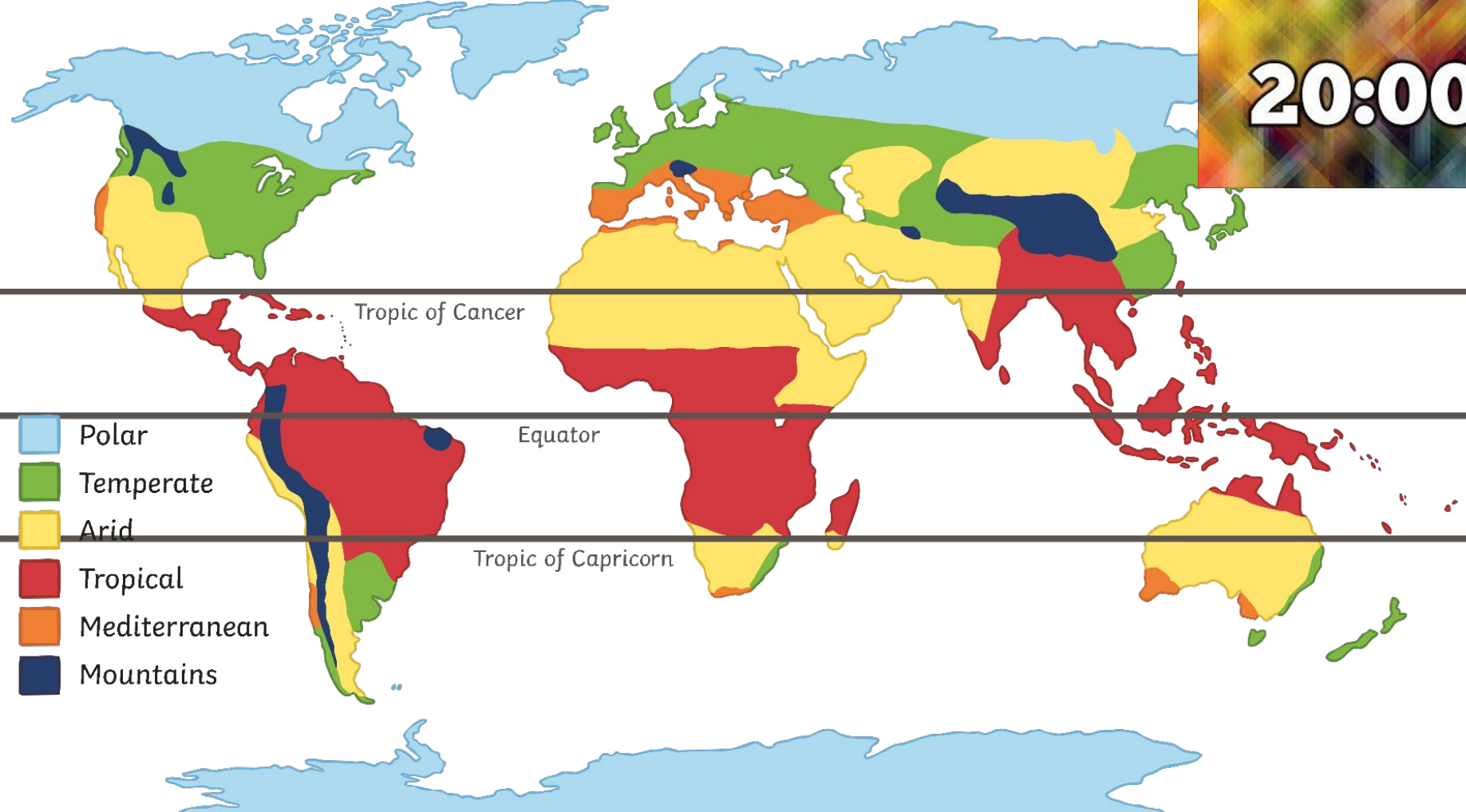
World Climate Zones
Shade in the different climate zones and complete the key. Add a short description of each climate zone in the boxes.

Polar Climate Zone		Temperate Climate Zone
Mountain Climate Zone		Arid Climate Zone
	☐ Polar ☐ Temperate ☐ Arid ☐ Tropical ☐ Mediterranean ☐ Mountains	
Mediterranean Climate Zone		Tropical Climate Zone

Secondary

visit www.visitniala.com

20:00



World Climate Zones

10:00

Complete the Which Climate Zone Activity Sheet.

You will need to search it up on iPad to locate each place and work out which climate zone it is in.



Google Maps

Which Climate Zone?

Use an atlas to work out which climate zone each location is in.

Polar	Temperate	Mediterranean	Arid	Tropical	Mountain

Alaska

Italy

Alps

Japan

Greenland

Saudi Arabia

Himalayas

Turkey

Ghana

Central Australia

Antarctica

Malaysia

Brazil

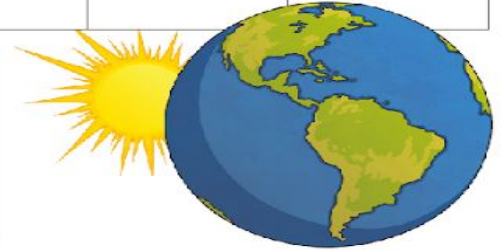
Poland

Chile

Egypt

Greece

UK



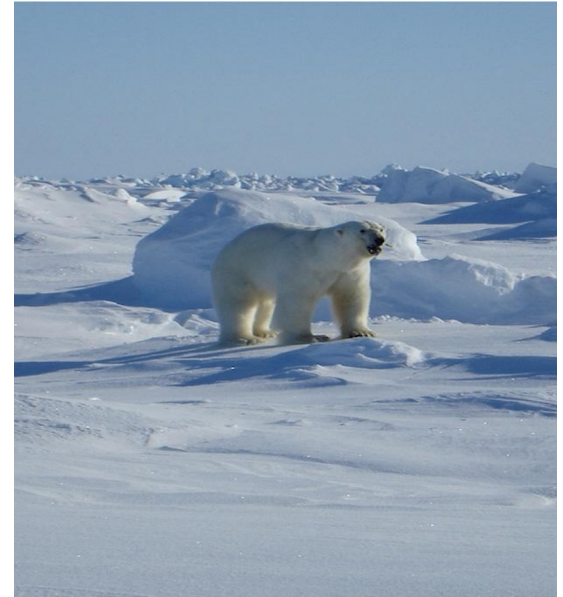
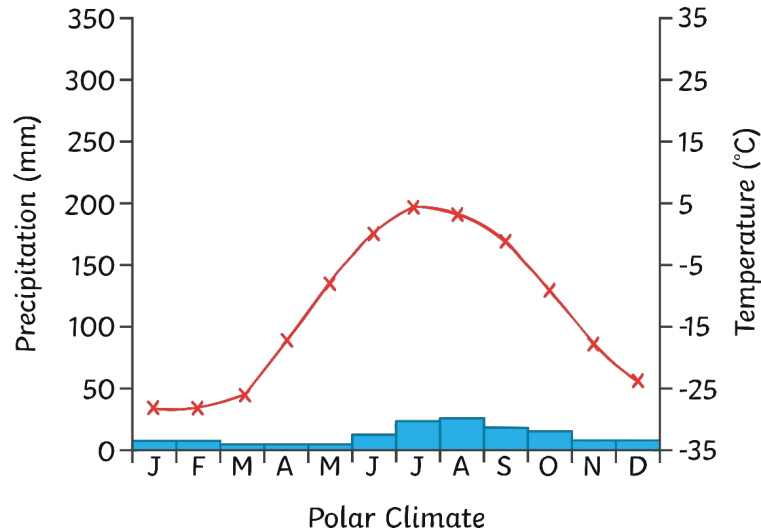
Secondary

visit twinkl.com



Crazy Climate Graphs!

polar climate zone

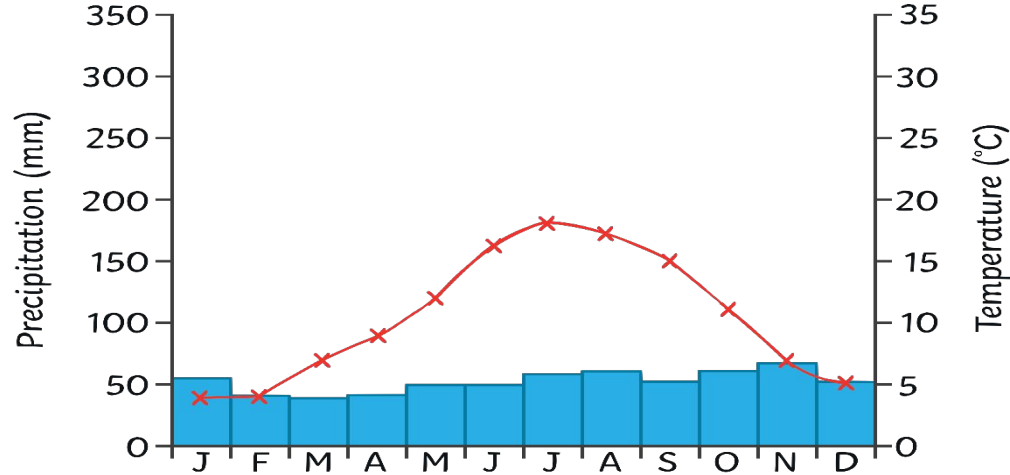


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Plenary

Crazy Climate Graphs!

temperate climate zone

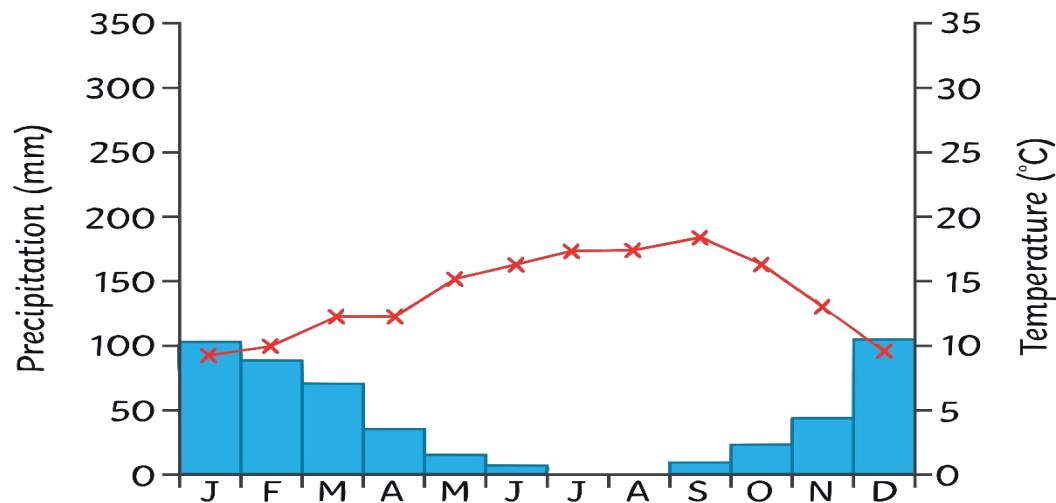


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Crazy Climate Graphs!

mediterranean climate zone

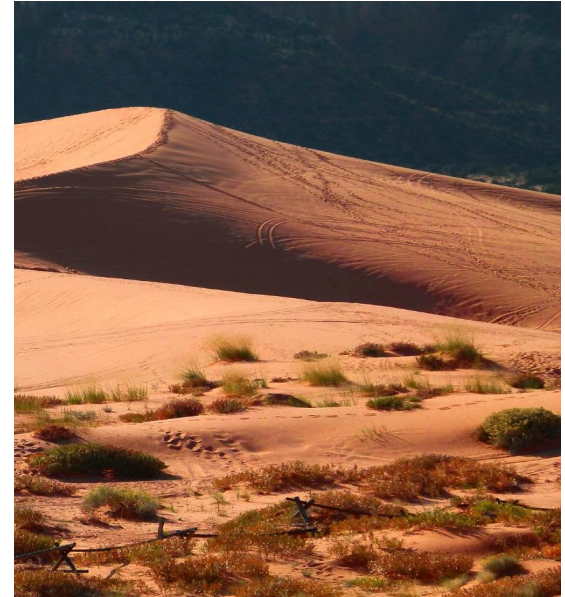
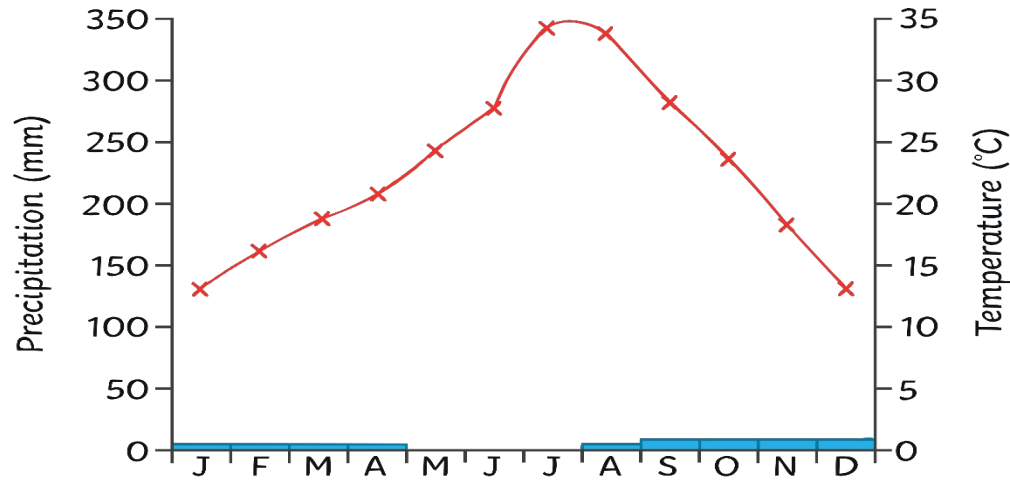


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Crazy Climate Graphs!

arid climate zone

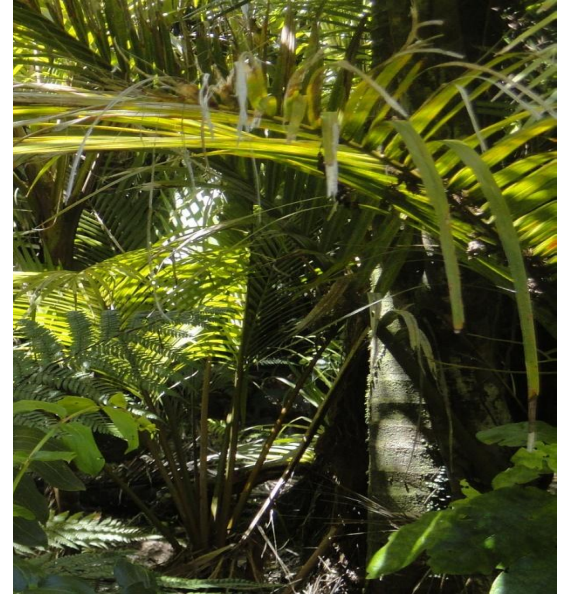
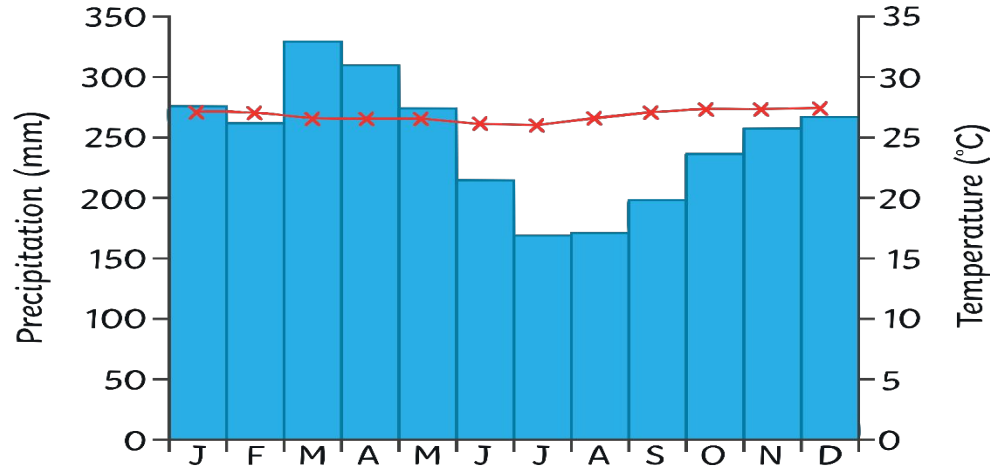


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Plenary

Crazy Climate Graphs!

tropical climate zone



Weather or Climate?

It was very snowy today:



Summers are warm here



It is has been cold here recently



It is very foggy



Winters are cold here



Droughts are common here



It is normally warm and dry here



But today
it rained
..... A lot !